

STUDYHUB
CLASS 12 — MP BOARD
PHYSICS — SET 1
ORIGINAL SAMPLE QUESTION PAPER

Time: 3 Hours	Maximum Marks: 70
Medium: English	Set: 1

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Internal options are given in each question from Question No. 6 to 20.
3. Questions 1 to 3 carry 6 marks each and Questions 4 and 5 carry 5 marks each. Each sub-question carries 1 mark.
4. Questions 6 to 12 carry 2 marks each. Answer within approximately 30 words.
5. Questions 13 to 16 carry 3 marks each. Answer within approximately 75 words.
6. Questions 17 to 20 carry 4 marks each. Answer within approximately 150 words.
7. Draw neat and labelled diagrams wherever necessary.

SECTION - A : OBJECTIVE TYPE QUESTIONS

Q.1 Choose and write the correct option. (1×6=6)

(i) The SI unit of electric flux is:

- A. N/C
- B. $\text{N m}^2/\text{C}$
- C. C/N
- D. $\text{N m}/\text{C}$

(ii) Two charges of equal magnitude and opposite sign separated by a small distance constitute:

- A. A monopole
- B. An electric dipole
- C. A capacitor
- D. A conductor

(iii) The resistance of a metallic conductor generally increases when:

- A. Temperature decreases
- B. Temperature increases
- C. Length decreases
- D. Area increases

(iv) The force on a charged particle moving parallel to a uniform magnetic field is:

- A. Maximum
- B. Minimum but non-zero
- C. Zero
- D. Infinite

(v) Electromagnetic waves are:

- A. Longitudinal waves
- B. Transverse waves
- C. Mechanical waves
- D. Sound waves

(vi) In a p-n junction diode under forward bias, the depletion layer:

- A. Becomes wider
- B. Becomes thinner
- C. Remains unchanged
- D. Disappears completely

Q.2 Fill in the blanks. (1×6=6)

(i) The electric field inside an electrostatic conductor is _____.

(ii) The SI unit of capacitance is _____.

(iii) The reciprocal of resistance is called _____.

(iv) The magnetic field at the centre of a circular current-carrying coil is proportional to _____.

(v) The phenomenon of production of induced current due to change in magnetic flux is called _____.

(vi) A semiconductor doped with a pentavalent impurity becomes an _____ type semiconductor.

Q.3 Write True or False. (1×6=6)

- (i) Electric field is a vector quantity.
- (ii) Equipotential surfaces can intersect each other.
- (iii) Kirchhoff's laws are based on conservation of charge and energy.
- (iv) A magnetic field can change the speed of a charged particle moving perpendicular to it.
- (v) Infrared radiation has a longer wavelength than visible light.
- (vi) A transformer works on the principle of mutual induction.

Q.4 Match Column A with Column B. (1×5=5)

Column A	Column B
(i) Electric potential	(a) Weber
(ii) Capacitance	(b) Ohm
(iii) Magnetic flux	(c) Volt
(iv) Resistance	(d) Farad
(v) Magnetic field	(e) Tesla

Q.5 Answer in one word/one sentence. (1×5=5)

- (i) What is the electric potential at infinity taken as?
- (ii) Write the SI unit of electric current.
- (iii) Which rule determines the direction of magnetic field around a straight current-carrying conductor?
- (iv) Name the phenomenon responsible for the working of an electric generator.
- (v) Which electromagnetic radiation is commonly used in remote controls?

SECTION - B : VERY SHORT ANSWER QUESTIONS

Q.6 (2 Marks)

Define electric dipole moment. Write its SI unit.

OR

State Gauss's law in electrostatics.

Q.7 (2 Marks)

What is an equipotential surface? Write one important property of it.

OR

Why is the electric field inside a conductor zero in electrostatic equilibrium?

Q.8 (2 Marks)

Two capacitors of $3\ \mu\text{F}$ and $6\ \mu\text{F}$ are connected in parallel. Find their equivalent capacitance.

OR

A capacitor of capacitance $5\ \mu\text{F}$ is charged to 20 V. Find the charge stored.

Q.9 (2 Marks)

State Ohm's law and write its mathematical expression.

OR

Why are household electrical appliances connected in parallel?

Q.10 (2 Marks)

A $10\ \Omega$ resistor is connected across 20 V. Calculate the current.

OR

Define drift velocity of free electrons.

Q.11 (2 Marks)

State Biot-Savart law.

OR

Why does a charged particle moving parallel to a magnetic field experience no magnetic force?

Q.12 (2 Marks)

Define electromagnetic induction.

OR

State Lenz's law.

SECTION - C : SHORT ANSWER QUESTIONS

Q.13 (3 Marks)

Derive the expression for electric field due to a point charge using Coulomb's law.

OR

Derive the relation between electric field and electric potential.

Q.14 (3 Marks)

A $6\ \Omega$ and a $3\ \Omega$ resistance are connected in parallel across a 12 V battery. Calculate the equivalent resistance and total current.

OR

Explain Kirchhoff's junction rule and loop rule.

Q.15 (3 Marks)

Derive the expression for force acting on a charged particle moving in a uniform magnetic field.

OR

Explain the working principle of a moving coil galvanometer.

Q.16 (3 Marks)

Explain interference of light and state the conditions necessary for sustained interference.

OR

What is diffraction of light? State one difference between diffraction and interference.

SECTION - D : LONG ANSWER QUESTIONS

Q.17 (4 Marks)

Derive the expression for capacitance of a parallel plate capacitor in air.

OR

Derive the expression for energy stored in a charged capacitor.

Q.18 (4 Marks)

Explain the principle, construction and working of an AC generator with a neat labelled diagram.

OR

Explain the working of a transformer and derive $V_s/V_p = N_s/N_p$.

Q.19 (4 Marks)

Using the lens formula, derive the relation between focal length, object distance and image distance for a thin lens.

OR

Explain Young's double-slit experiment and obtain the expression $\beta = \lambda D/d$.

Q.20 (4 Marks)

Explain the working of a p-n junction diode under forward and reverse bias with suitable diagrams. Also explain the depletion region.

OR

Explain the working of a transistor as an amplifier with a suitable circuit diagram.

MARKS VERIFICATION

Question	Marks
Q.1	6
Q.2	6
Q.3	6
Q.4	5
Q.5	5
Q.6-Q.12	14
Q.13-Q.16	12
Q.17-Q.20	16
TOTAL	70

6 + 6 + 6 + 5 + 5 + 14 + 12 + 16 = 70 Marks ✓

SHORT ANSWER KEY

Q.1: B, B, B, C, B, B. Q.2: zero; farad; conductance; current; electromagnetic induction; n-type. Q.3: True, False, True, False, True, True. Q.4: (i)-(c), (ii)-(d), (iii)-(a), (iv)-(b), (v)-(e). Q.5: zero; ampere; right-hand thumb rule; electromagnetic induction; infrared radiation.

For Q.6-Q.20, award marks for correct formulae, reasoning, derivation, calculation, diagram and final answer as applicable. Full solutions are not included.